

BMS SYSTEM HEALTH REPORT

Building A (Siemens DesigoCC) & Building B (Niagara Tridium)

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01 Executive Summary

This report presents the findings of a routine system analysis conducted on the Building Management Systems of two independently operated buildings located on the same site — referred to as **Building A** (managed on Siemens DesigoCC) and **Building B** (managed on Niagara Tridium). Both buildings share a centralised chiller plant located on the roof.

The analysis covered system architecture review, active alarm and fault log assessment, and identification of operational risks requiring attention. It was conducted by the site automation engineer as part of ongoing monitoring responsibilities and is intended for review by the building manager.

Building A operates across Ground to Fourth Floor, with 5 AHUs per floor distributing conditioned air via VAVs and FCUs. At the time of this report, the system has a few active alarms and faults related to equipment that has been identified for repair, currently pending approval or procurement.

Building B operates across First to Fourth Floor, with 5 AHUs per floor distributing conditioned air via diffuser units. The system is operational with no critical alarms reported at this time.

Shared infrastructure — the roof-level chiller plant — currently shows 2 Life Safety alarms that have been logged and are pending action by the mechanical contractor via the site manager.

02 System Overview

Parameter	Building A	Building B
BMS Platform	Siemens DesigoCC	Niagara Tridium
Floors Served	Ground + Floors 1–4 (5 floors)	Floors 1–4 (4 floors)
AHUs per Floor	5	5
Total AHUs	25	20
Distribution System	VAVs and FCUs	Diffuser units with temp sensors
AHU Components	VSD fan, chilled water valve, filters, heating interlock, supply/return air temp, static pressure, CHW supply/return temp sensors, time clock	VSD fan (SAF), chilled water valve, Heater Step 1 & 2, supply/return air temp, CHW supply/return temp, static pressure sensors
Controllers	Demand Controllers	MLM Controllers + Rickard PSUs
Communication	Independent BACnet network	Independent BACnet network
Monitoring	Daily — DesigoCC SCADA	Daily — Niagara SCADA

Reporting	Monthly service report to building manager	Monthly service report to building manager
Shared Infrastructure	Roof-level chiller plant (serves both buildings)	Roof-level chiller plant (serves both buildings)

03 Alarm & Fault Summary

The following summarises the current alarm and fault status across both buildings as observed on the Desigo CC platform on 1 June 2026. Building B had no active alarms at the time of review.

Category	Count	Description	Building	Status
Life Safety	2	Chiller alarms at roof level — East Side Chiller Plantroom	Shared	Pending — mechanical contractor
High	1	Room temperature high limit (22°C) — Floor 4, AHU 19	Building A	Active — pending repair
Medium	282	Room temperature high limit alarms across multiple floors and AHUs	Building A	Active — equipment pending approval/procurement
Fault	1,450	Multiple faults on Floor 2 AHU 15 — room temp setpoint, heating enable, damper output, HVAC mode, start command	Building A	Active — equipment pending approval/procurement
Information	3	Alarm suppression active; system information events	Building A	Noted
Building B	0	No active alarms at time of review	Building B	Normal

04 Key Observations

Chiller Plant (Shared — Roof Level):	Two Life Safety alarms are active on the East Side Chiller Plantroom. These have been escalated to the site manager for action by the mechanical contractor. Both buildings are affected by any chiller unavailability.
Building A — Room Temperature Pattern:	A significant number of medium-level room temperature high limit alarms are active across multiple floors. The pattern across floors and AHU zones suggests a cooling performance issue that goes beyond isolated equipment faults. This is being monitored and relevant equipment has been identified for repair pending approval or procurement.
Building A — AHU 15 (Floor 2):	AHU 15 on the Second Floor is showing multiple simultaneous active faults including controller HVAC mode, heating enable, damper output, room temperature setpoint, and start command. This unit requires physical inspection and repair. It is currently pending approval or procurement of the necessary parts or equipment.
Building B — Operational:	Building B is operating normally across all four floors. All diffuser units, AHUs, MLM controllers, and Rickard PSUs are functioning within expected parameters. No alarms are active at the time of this report.

Priority	Item	Action	Responsible
Immediate	Chiller — Life Safety Alarms	Mechanical contractor to inspect and resolve chiller alarms at roof level	Site Manager / Mechanical Contractor
Pending Approval	Building A — AHU 15 (Floor 2)	Procure and replace or repair faulty components; resolve controller faults	Site Automation Engineer / Building Manager
Pending Approval	Building A — Room Temp Alarms (multiple floors)	Identify and repair equipment contributing to temperature exceedances floor by floor	Site Automation Engineer / Building Manager
Ongoing	Building B — Routine Monitoring	Continue daily monitoring; no corrective action required at this time	Site Automation Engineer